

SAMRUDDHI KAULE

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AI/ML Developer | Full-Stack Developer

SUMMARY

Final-year Computer Science undergraduate with a strong foundation in Artificial Intelligence and Machine Learning hands-on experience in data-driven problem solving, machine learning workflows, application development, and research, supported by practical technical work. Analytical, adaptable, and self-driven, with a strong ability to learn new technologies and translate concepts into practical solutions.

SKILLS

Programming Languages: Python, SQL, C, C++

Machine Learning & AI: PyTorch, Scikit-learn, NLP, Computer Vision, Recommendation Systems

Data & Analytics: Pandas, NumPy, Feature Engineering, Power BI

Frameworks & Tools: FastAPI, Django, OpenCV, Git

Web & Databases: React, REST APIs, PostgreSQL, MongoDB

PROJECTS

Real-Time Hand Gesture Recognition System [GitHub](#)

Developed a real-time gesture recognition system classifying 36 alphabet/digit gestures from 42 normalized hand-landmark features, achieving 95.87% accuracy and 0.952 Macro F1 on 653 held-out samples.

Built a position- and scale-invariant feature normalization pipeline and trained a Random Forest ensemble on 3,263 samples, achieving a 91.27 percentage-point lift over the majority-class baseline.

Optimized inference for low-latency deployment, achieving 3.21 ms median and 5.58 ms P95 latency at 275 predictions/second on precomputed landmarks.

CineMatch ? Hybrid Movie Recommendation System [GitHub](#)

Built a hybrid recommender using TF-IDF (Scikit-learn), SVD (Surprise), and NCF (PyTorch) to address cold-start and sparsity.

Added context-aware personalization based on user mood and time-of-day, and used FAISS for sub-millisecond real-time similarity search.

Developed a FastAPI backend with a React frontend and NLP-based chatbot for interaction; evaluated using RMSE (~1.04), Precision@10, and Hit Ratio@10, improving over standalone models.

RetailX ? Sales Analytics [Github](#)

Built an end-to-end sales analytics pipeline processing 32K+ transactions, 20K orders, 2,999 customers, and 500 products using Python, SQL, ML, and Power BI.

Cleaned and validated data by removing 159 duplicates, resolving 966 missing values, and correcting 65 extreme discounts, retaining 99.51% of records.

Developed K-Means customer segmentation using 6 behavioral features, identifying 4 customer segments, including one segment representing 23.27% of customers and 38.01% of revenue.

PROFESSIONAL EXPERIENCE

AI-ML Virtual Intern ? Google for Developers (EduSkills) Apr 2025 ? Jun 2025

Performed data preprocessing, feature engineering, and model training on real-world datasets using Python.

Applied machine learning techniques for classification and prediction tasks, evaluating performance using standard metrics.

Implemented end-to-end ML workflows including data cleaning, model building, and result

interpretation.

Data Science Virtual Internship ? Siemens (EduSkills) Apr 2026 ? Jun 2026

Completed a 10-week virtual internship covering data science, machine learning, and data analytics concepts.

Used the RapidMiner platform for data preprocessing, visualization, model building, and evaluation through hands-on assessments.

EDUCATION

MIT-ADT University, MIT School of Computing Expected May 2027

B.Tech in Computer Science (AI & Edge Computing) ? CGPA: 8.47 / 10

CERTIFICATIONS & ADDITIONAL INFORMATION

Machine Learning I (Columbia) ? AI Agents (Google, via Kaggle) ? AI Principles with Edge Computing (Coursera) ? Power BI Bootcamp (Nucleus, SSCBS)

Languages: English, Hindi, Marathi, Basic German

Interests: Basketball, Tennis, Reading